

1. Output Screenshots

- Normal Configuration: Accurate Inputs

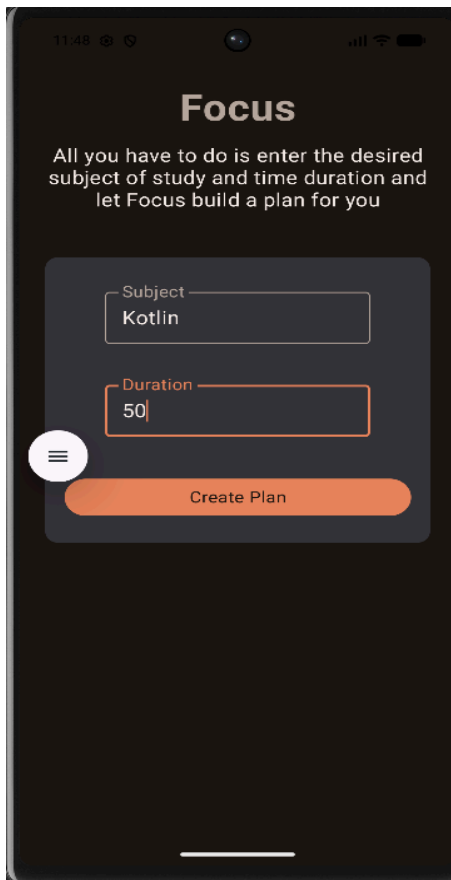


Fig 1: Entering Input

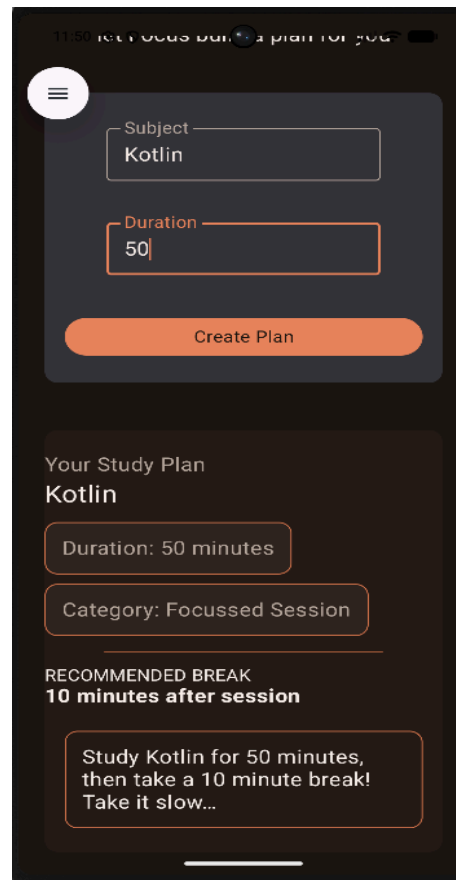


Fig 2: Generated Study Plan

- Invalid Inputs

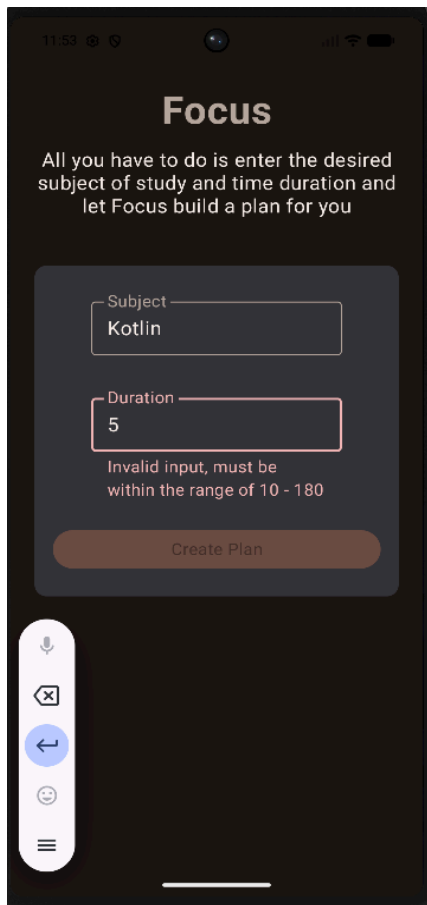


Fig 3: Invalid Duration

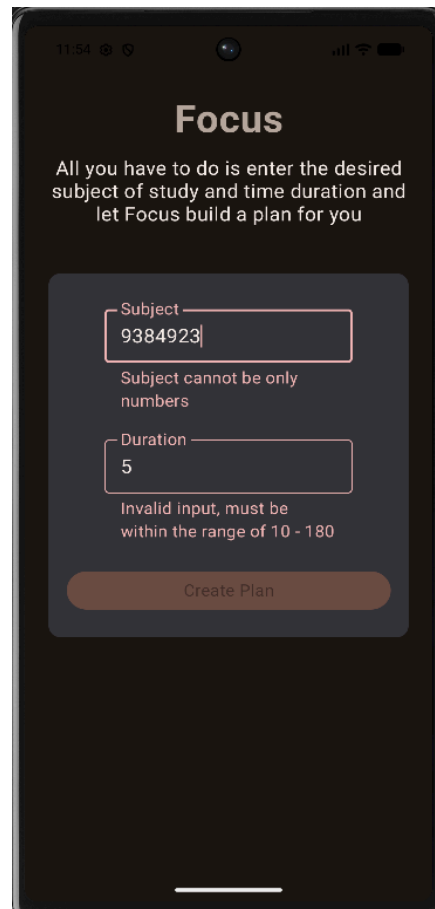


Fig 4: Invalid Subject

- Screen Rotation

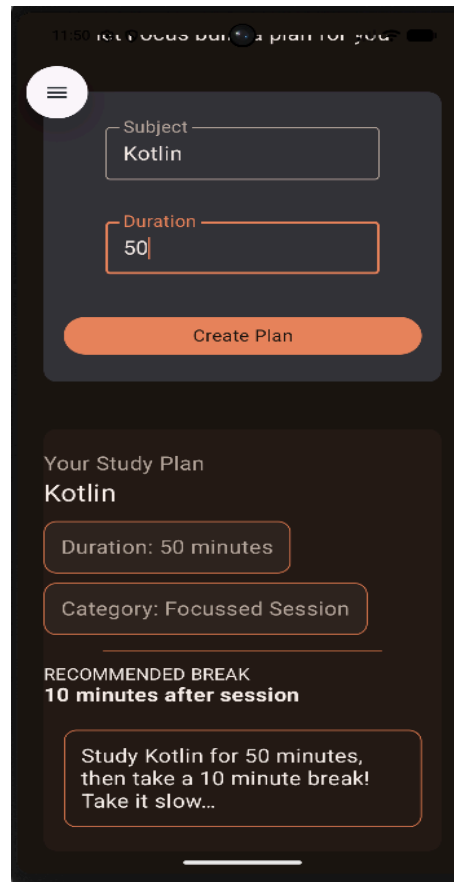


Fig 5: Before Rotation

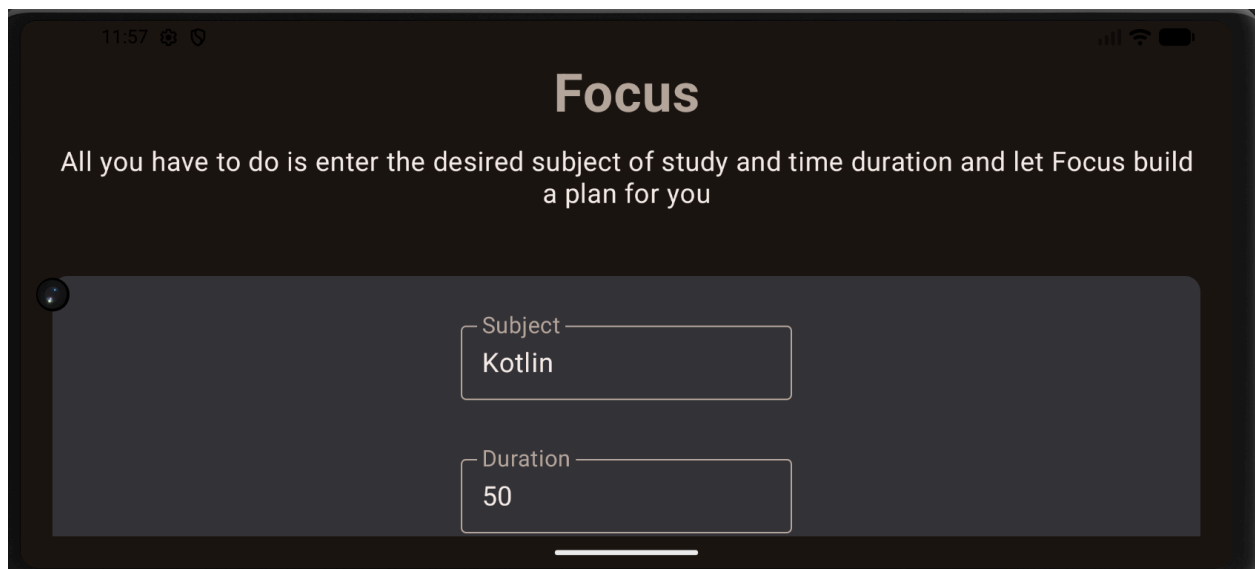


Fig 6: After Rotation

2. Questions

Q. Which composable owns the application state?

Ans. The composable 'FocusPlanRoute' owns the application state i.e the subject, minutes and, the plan.

Q. Why are the text-field values stored as String rather than Int?

Ans. The text-field always works with String values since it is not natively built to support integer values.

Q. Why is toIntOrNull() safer than toInt() for this application?

Ans. If the user left the field blank or entered a non-numeric value, and the toInt() conversion took place, the application would crash. Hence, it is safer to use toIntOrNull().

Q. What state change causes the button to be recomposed?

Ans. When the field for either state or the duration changes, specifically the canCreatePlan value, the button is recomposed since the button is enabled only when both the inputs are valid (determined by canCreatePlan).

Q. What does rememberSaveable preserve that a local variable would not?

Ans. rememberSaveable saves the value into the Activity's saved instance state Bundle, so it survives configuration changes like rotation. A local variable or one using only remember is lost when the Activity is destroyed and recreated, which resets the app to its initial value/state.